Loops

while Loop

for Loop

do while Loop

while Loop

```
while(condition) {

//do something

}
```

JavaBasics.java 1

JavaBasics.java > JavaBasics > main(String[])

APNA COLLEGE

```java
import java.util.*;

public class JavaBasics {
    Run | Debug
    public static void main(String args[]) {
        int counter = 0;
        while(counter < 100) {
            System.out.println("Hello World");
            counter++;
        }

    }
}

```

JavaBasics.java 1

JavaBasics.java > JavaBasics > main(String[])

```java
import java.util.*;

public class JavaBasics {
    Run | Debug
    public static void main(String args[]) {
        int counter = 0;
        while(counter < 100) {
            System.out.println("Hello World");
            counter++;
        }

        System.out.println("printer HW 100x");
    }
}

```

Print number from 1 to 10

1 2 3 4 5 6 7 8 9 10 → while

int counter= 1 ;

```java
public class JavaBasics {
    Run | Debug
    public static void main(String args[]) {
        int counter = 1;
        while(counter <= 10) {
            System.out.println(counter);
            counter++;
        }
    }
}
```

PROBLEMS 1   OUTPUT   TERMINAL   DEBUG CONSOLE   Filter (e.g. text, **/*.ts, !**/nod

JavaBasics.java 1

⚠ The import java.util is never used Java(268435844) [1, 8]

user (sc)

Print number from 1 to n

int n = range

n = 7

```
int counter = 1;
while ( counter <= n) {

}
```

user input

JavaBasics.java > JavaBasics > main(String[])


```java
public class JavaBasics {
    Run | Debug
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        int range = sc.nextInt();
        int counter = 1;

        while(counter <= range) {
            System.out.print(counter + " ");
            counter++;
        }
        System.out.println();

```

PROBLEMS 1 OUTPUT TERMINAL DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.ja
25
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25
shradhakhapra@Shradhas-Air Java Programming Language %
```

for Loop

```
for(initialisation; condition; updation) {

//do something

}
```

APNA COLLEGE

```java
public class JavaBasics {
    Run | Debug
    public static void main(String args[]) {
        // int i=1;
        for(int i=1; i<=100; i++) {
            System.out.println("Hello World");
        }
    }
}
```

PROBLEMS 1 OUTPUT TERMINAL DEBUG CONSOLE

```
Hello World
Hello World
Hello World
Hello World
Hello World
Hello World
Hello World
Hello World
Hello World
Hello World
shradhakhapra@Shradhas-Air Java Programming Language %
```

Print SQUARE pattern

```
* * * * → L1        4 stars
* * * * → L2
* * * * → L3
* * * * → L4
```

```java
public class JavaBasics {
    Run | Debug
    public static void main(String args[]) {
        for(int line=1; line<=4; line++) {
            System.out.println("****");
        }
    }
}
```

PROBLEMS 1 OUTPUT TERMINAL DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
****
****
****
****
shradhakhapra@Shradhas-Air Java Programming Language %
```


Print Reverse of a number

n = 10899 → 99801

Print Reverse of a number

n = 10899

%

number = 132/10

↓

remainder = lastDigit

10) 132 (13
 − 130
 2

Print Reverse of a number

n = 10899

number = 132/10

53/10

5

remainder = lastDigit

10/10

1

9

lastDigit = n%10;

JAVA Programming Language

① last Dig num%10

② remove last Dig num/10

APNA COLLEGE

Print Reverse of a number

n = 10899

lastDigit = n%10 ;

9

JAVA Programming Language

Print Reverse of a number

1. last Dig $num\%10$
2. remove last Dig $num/10$

APNA COLLEGE

n = 10899

99801

```
lastDigit = n%10;

n = n/10;
```

(1089)

(108)

(10)

(1)

(0)

JAVA Programming Language

Print Reverse of a number

① last Dig   num%10

② remove last Dig   num/10

n = 10899

```
while (n > 0) {
    lastDigit = n % 10;
    n = n/10;
```

(1089)
(108)
(10)
(1)
(0)

99801

APNA COLLEGE

Reverse the given number

n = 10899

int reverse =

rev = (rev * 10) + last Digit

last Digit

$rev = (rev * 10) + lastDigit$

APNA COLLEGE

Reverse the given number

n = 10899

$lD = 9$

$n = 1089$

108

$rev = 0$

$rev = (0 * 0) + 9 = 9$

$rev = (9 * 10) + 9 = 90 + 9 = 99$

$rev = (99 * 10) + 8 = 990 + 8 = 998$

JAVA Programming Language

$rev = (rev * 10) + lastDigit$

Reverse the given number

n = 10899

$LD = 9$

$n = 1089$

108

$rev = 0$

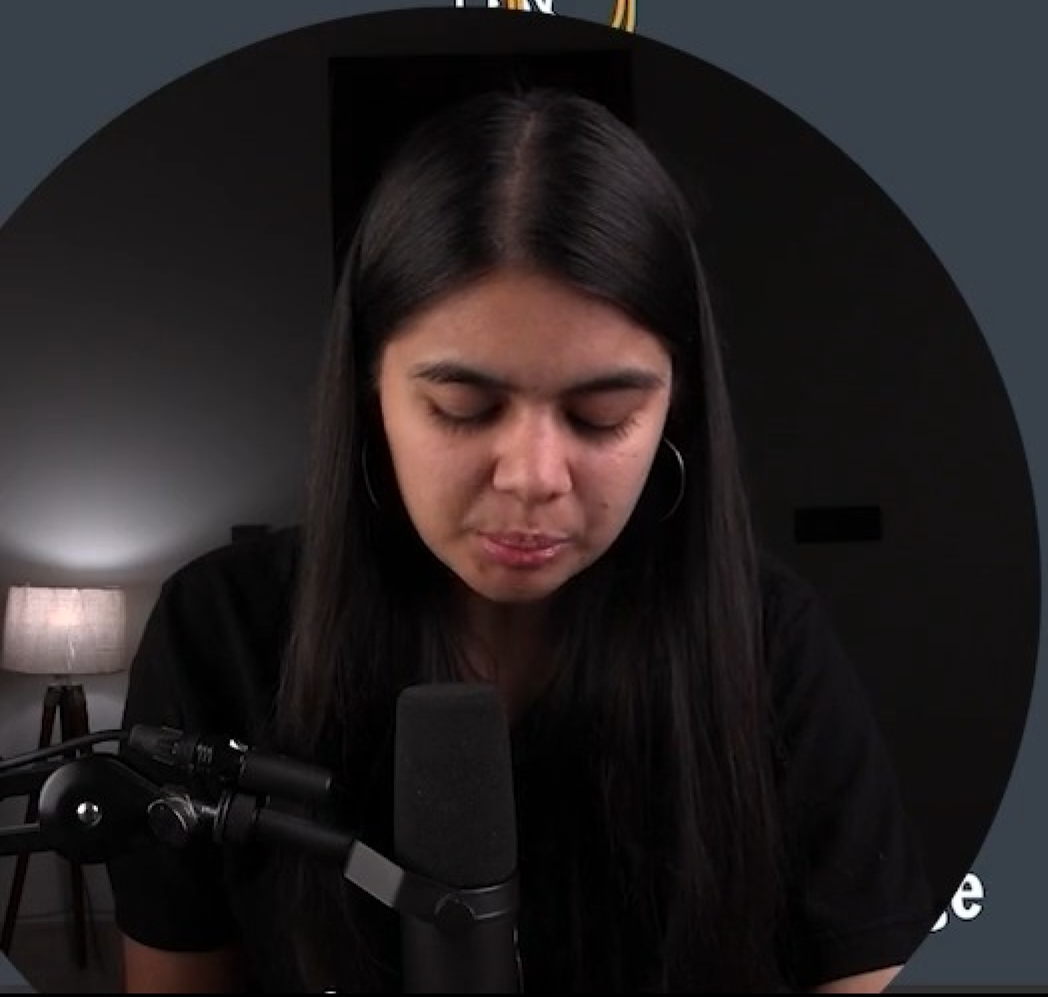
$rev = (0 * 0) + 9 = 9$

$rev = (9 * 10) + 9 = 90 + 9 = 99$

$rev = (99 * 10) + 8 = 990 + 8 = 998$

$rev = (998 * 10) + 0 = 9980$

$rev = (9980 * 10) + 1 = 99800 + 1 = 99801$



Reverse the given number

APNA COLLEGE

rev = (rev * 10) + lastDigit

rev = 0
rev = (0*0) + 9 = 9
rev = (9*10) + 9 = 90+9 = 99
rev = (99 *10) + 8 = 990+8 = 998
rev = (998 *10) + 0 = 9980
rev = (9980*10) + 1 = 99800+1 = 99801

ones
2nd
last

n = 10899

ones

tens

LD = 9

n = 1089
108
10
1

```
int rev = 0 ;

while ( n > 0 ) {
    lastDigit
    rev = rev * 10 + LD
    n update
}
print(rev)
```


JAVA Programming Language

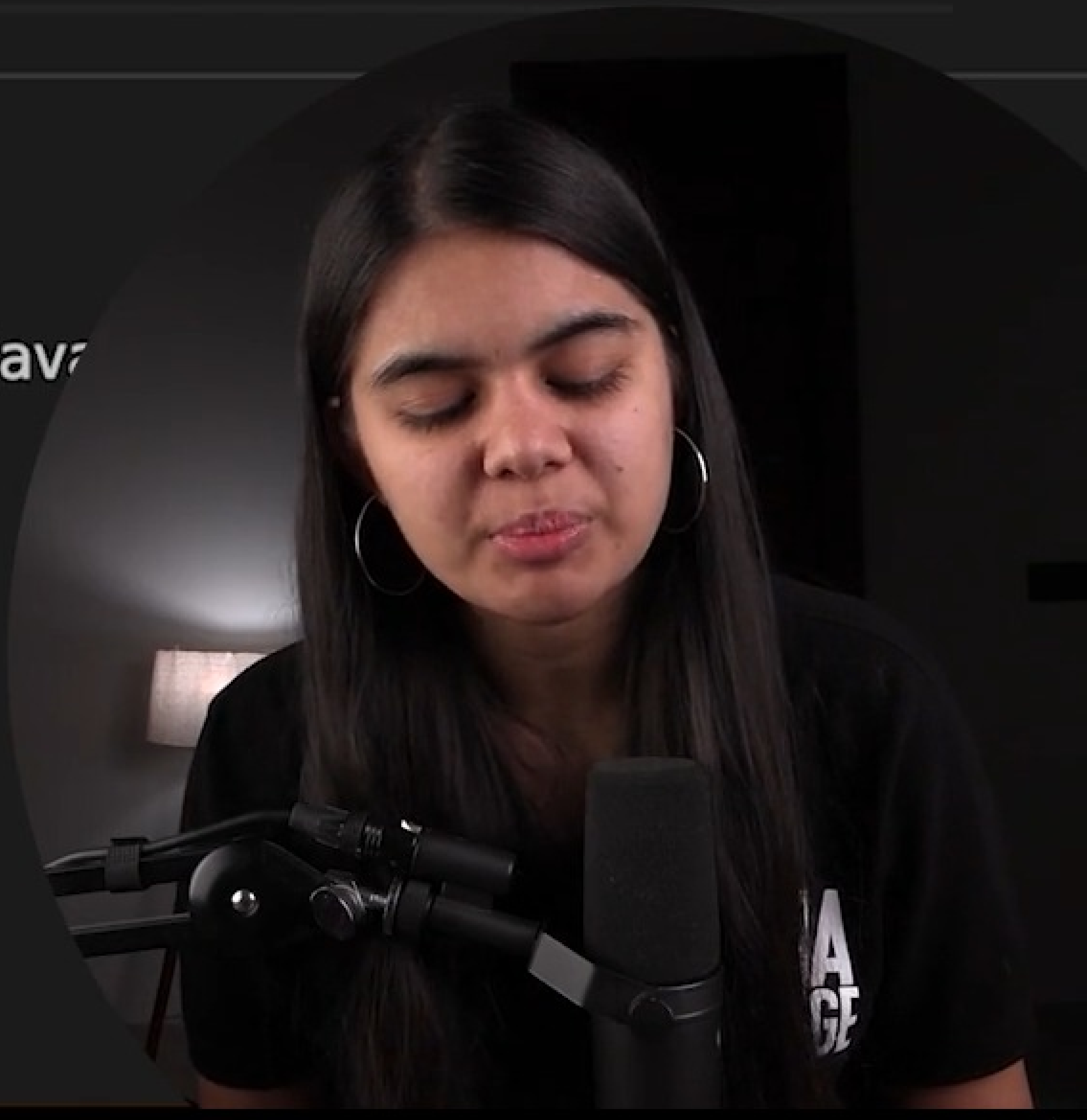
```java
    public static void main(String args[]) {
        int n = 10899;
        int rev = 0;

        while(n > 0) {
            int lastDigit = n % 10;
            rev = (rev*10) + lastDigit;
            n = n/10;
        }

        System.out.println(rev);
    }
```

PROBLEMS 1 OUTPUT TERMINAL DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
99801
shradhakhapra@Shradhas-Air Java Programming Language %
```



armankhan082020@gmail.com

JavaBasics.java 1

JavaBasics.java > JavaBasics > main(String[])

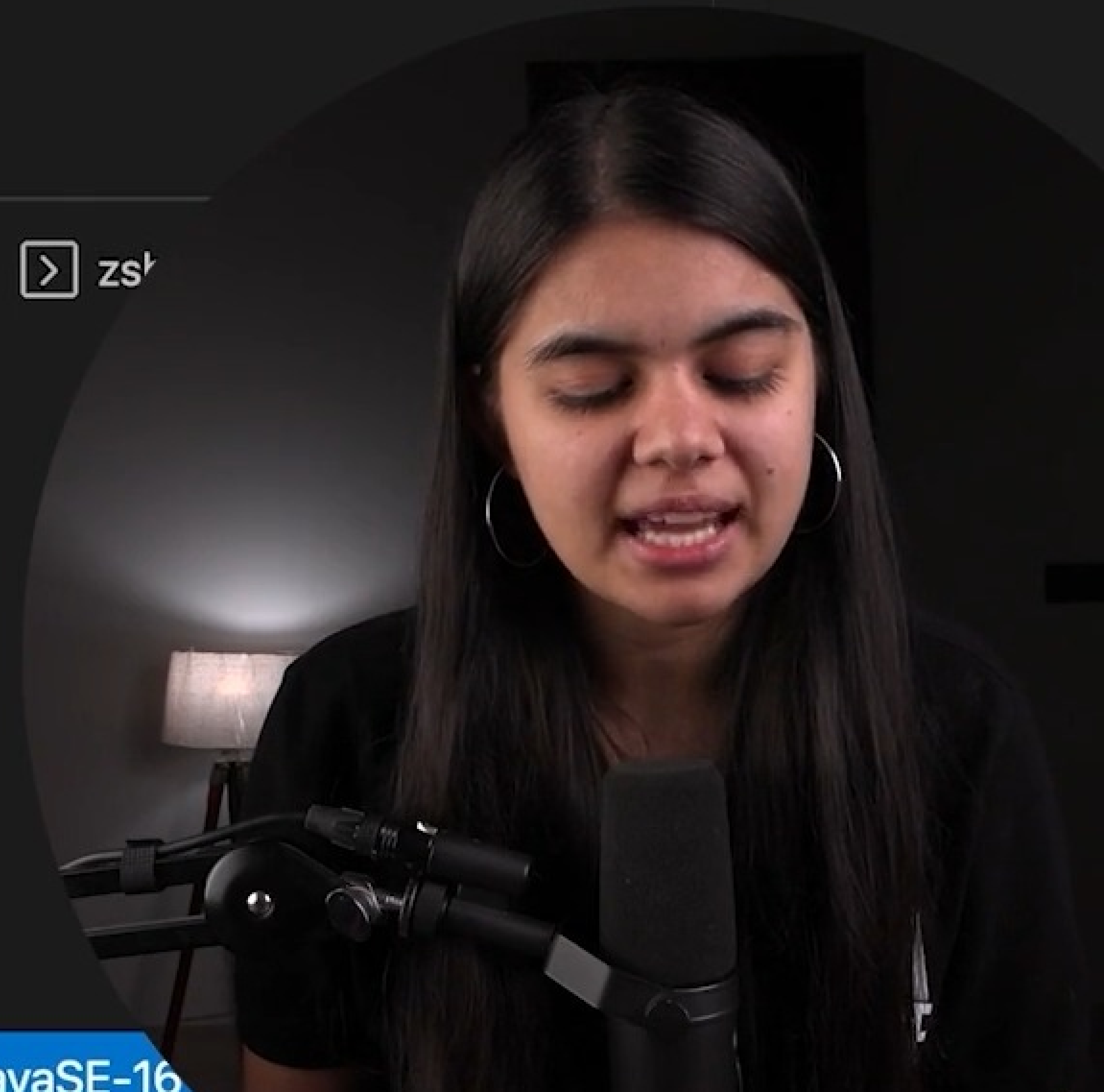
```java
import java.util.*;

public class JavaBasics {
    public static void main(String args[]) {
        int counter = 1;
        do {
            System.out.println("Hello World");
            counter++;
        } while(counter <= 10);
    }
}
```

PROBLEMS 1 OUTPUT TERMINAL DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
Hello World
Hello World
Hello World
Hello World
Hello World
Hello World
Hello World
Hello World
Hello World
Hello World
```

Run Testcases 0 1 Spaces: 4 UTF-8 LF Java Go Live JavaSE-16



APNA COLLEGE

Break Statement

to exit the loop

JAVA Programming Language

armankhan082020@gmail.com

JavaBasics.java 1

JavaBasics.java > JavaBasics > main(String[])

```java
import java.util.*;

public class JavaBasics {
    Run | Debug
    public static void main(String args[]) {
        for(int i=1; i<=5; i++) {
            if(i == 3) {
                break;
            }
            System.out.println(i);
        }

        System.out.println("i am out of the loop");
```

PROBLEMS 1   OUTPUT   TERMINAL   DEBUG CONSOLE   Filter (e.g. text, **/*.ts, !**/no

JavaBasics.java 1

⚠ The import java.util is never used Java(268435844) [1, 8]

JavaBasics.java 1

JavaBasics.java > JavaBasics > main(String[])

Run | Debug

```java
public class JavaBasics {
    public static void main(String args[]) {
        for(int i=1; i<=5; i++) {
            if(i == 3) {
                break;
            }
            System.out.println(i);
        }

        System.out.println("i am out of the loop");
    }
}
```

PROBLEMS 1 OUTPUT TERMINAL DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
1
2
i am out of the loop
shradhakhapra@Shradhas-Air Java Programming Language %
```

JavaBasics.java 1

JavaBasics.java > JavaBasics > main(String[])


```java
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);

        do {
            System.out.print("enter your number : ");
            int n = sc.nextInt();

            if(n % 10 == 0) {
                break;
            }
            System.out.println(n);
        } while(true);
    }
```

PROBLEMS 1 OUTPUT TERMINAL DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
enter your number : 4
4
enter your number : 
```

Continue Statement

to skip an iteration

1 to ~~n~~ 5

1 2 3 4 5

i==3

continue

JavaBasics.java 1

JavaBasics.java > JavaBasics > main(String[])

```java
import java.util.*;

public class JavaBasics {
    public static void main(String args[]) {
        for(int i=1; i<=5; i++) {
            if(i == 3) {
                continue;
            }
            System.out.println(i);
        }
    }
}
```

PROBLEMS 1 OUTPUT TERMINAL DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
1
2
4
5
shradhakhapra@Shradhas-Air Java Programming Language % 
```

Display all numbers entered by user except multiples of 10

1	1
5	5
10	
7	7
50	
25	25

JavaBasics.java 1 ×

JavaBasics.java > JavaBasics > main(String[])

```java
import java.util.*;

public class JavaBasics {
    Run | Debug
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);

        do {
            System.out.print("enter your number : ");
            int n = sc.nextInt();

            if(n % 10 == 0) {
                continue;
            }

            System.out.println("number was : " + n);
        } while(true);

    }
}
```

```java

        do {
            System.out.print("enter your number : ");
            int n = sc.nextInt();

            if(n % 10 == 0) {
                continue;
            }

            System.out.println("number was : " + n);
        } while(true);

    }
```

PROBLEMS 1 OUTPUT TERMINAL DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
enter your number : 5
number was : 5
enter your number : 9
number was : 9
enter your number : 20
enter your number :
```

Check if a number is prime or not

prime : 2, 3, 5, 7, 11

Composite

PRIME : $n = a \times b$

$n \leftarrow$ user

prime : $n = 1 \times n$, $n \times 1$

$4 = 1 \times 4$, 2×2, 4×1

$6 = 1 \times 6$, 2×3, 3×2, 6×1

$9 = 1 \times 9$, 3×3, 9×1

$12 = 1 \times 12$, 2×6, 3×4, 4×3, 6×2, 12×1

$$\boxed{n = \sqrt{n} \times \sqrt{n}}$$

1 to $\sqrt{n}$

APNA COLLEGE

Check if a number is prime or not

prime : 2, 3, 5, 7, 11

$n = a \times b$

$n = i \times b$

2 to n-1

$\boxed{12}$ 2 to 11

(n)

$12 = 1 \times 12$
2×6
3×4
4×3
6×2
12×1

12 % 4 == 0
12 % 3 == 0
12 % 1 == 0
12 % 12 == 0
12 % 2 == 0
12 % 6 == 0

$$\boxed{n = \sqrt{n} \times \sqrt{n}}$$

JAVA Programming Language

APNA COLLEGE

Check if a number is prime or not

prime : 2, 3, 5, 7, 11

$n = a \times b$

$\underline{n = i \times b}$

2 to n-1

```
for(int i=2; i<=n-1; i++)
    n%i==0 X
```

prime

$n = i \times b$

$$\boxed{n = \sqrt{n} \times \sqrt{n}}$$

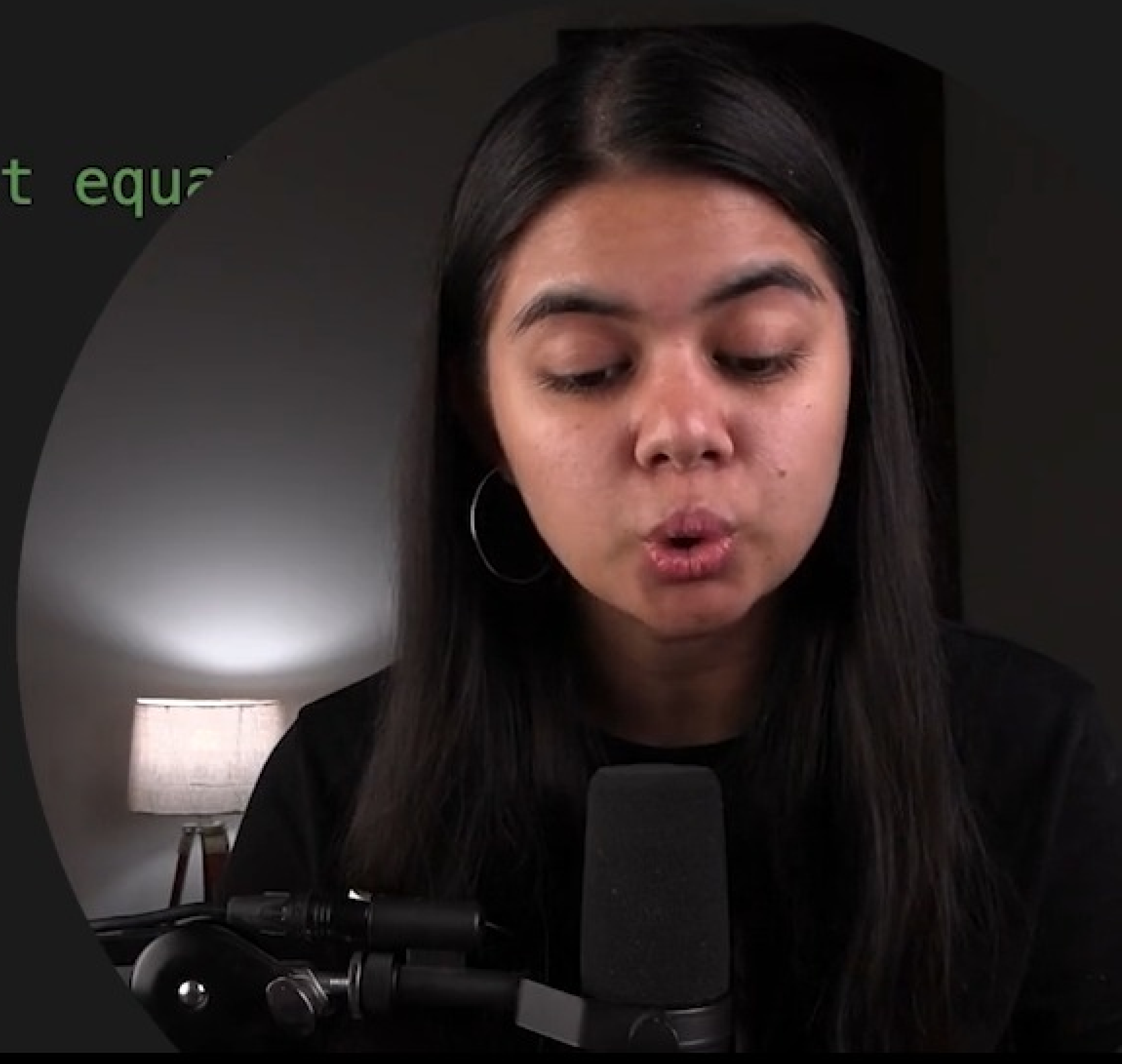
JAVA Programming Language

JavaBasics.java > JavaBasics > main(String[])

```java
public class JavaBasics {
    Run | Debug
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        if(n == 2) {
            System.out.println("n is prime");
        } else {
            boolean isPrime = true;
            for(int i=2; i<=n-1; i++) {
                if(n % i == 0) { // n is a multiple of i (i not equa
                    isPrime = false;
                }
            }

            if(isPrime == true) {
                System.out.println("n is prime");
            } else {
                System.out.println("n is not prime");
            }
```



$n - 2$ times ✓

Check if a number is prime or not

prime : 2, 3, 5, 7, 11

12 :

$$\left.\begin{array}{l} 1 \times 12 \\ 2 \times 6 \\ 3 \times 4 \end{array}\right\} \text{unique}$$

$$\left.\begin{array}{l} 4 \times 3 \\ 6 \times 2 \\ 12 \times 1 \end{array}\right\} \text{repeat}$$

10 :

$$\left.\begin{array}{l} 1 \times 10 \\ 2 \times 5 \end{array}\right\} \text{unique}$$

$$\left.\begin{array}{l} 5 \times 2 \\ 10 \times 1 \end{array}\right\} \text{repeat}$$

$$\boxed{n = \sqrt{n} \times \sqrt{n}}$$

$n-2$ times ✓

Check if a number is prime or not

prime : 2, 3, 5, 7, 11

for(int i=2 ~~to~~ $\sqrt{n}$)

$i <= \sqrt{n}$

$\boxed{n = \sqrt{n} \times \sqrt{n}}$

12 :
1×12
2×6
3×4 } unique
4×3
6×2
12×1 } repeat

10 :
1×10
2×5 } unique
5×2
10×1 } repeat

JAVA Programming Language

Check if a number is prime or not

6

7

1000000

$n-2$ times ✓

$\sqrt{n}$ times

prime : 2, 3, 5, 7, 11

$\text{for(int } i=2 \text{ } \cancel{\text{to}} \text{ } \sqrt{n})$

$i <= \sqrt{n}$

12 :

1×12
2×6
3×4 } unique

4×3
6×2
12×1 } repeat

10 :

1×10
2×5 } unique

5×2
10×1 } repeat

$\sqrt{n} < (n-2)$

$$\boxed{n = \sqrt{n} \times \sqrt{n}}$$

JAVA Programming Language

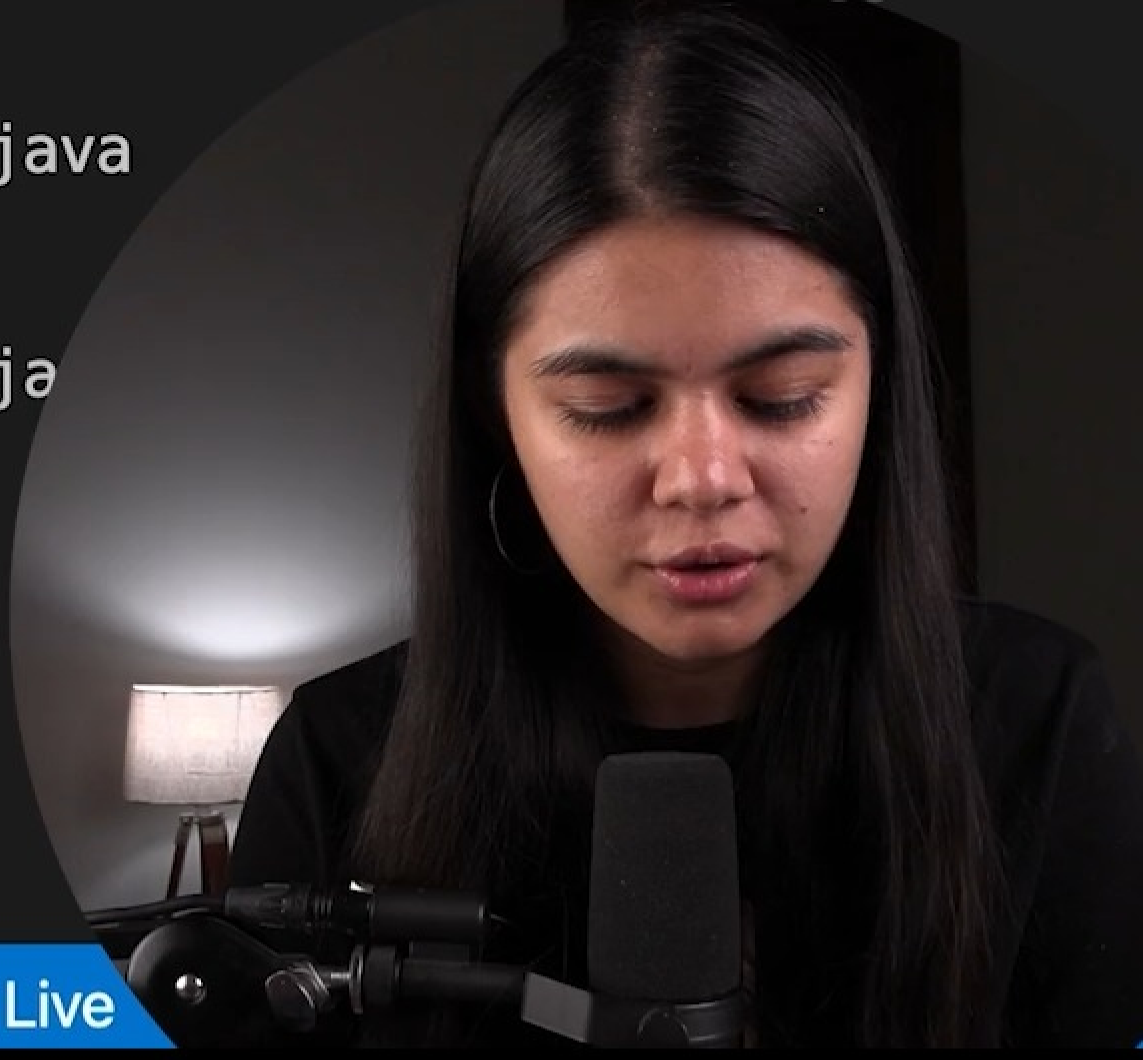
APNA COLLEGE

```java
        if(n == 2) {
            System.out.println("n is prime");
        } else {
            boolean isPrime = true;
            for(int i=2; i<=Math.sqrt(n); i++) {
                if(n % i == 0) { // n is a multiple of i (i not equal to 1 or n)
                    isPrime = false;
                }
            }

            if(isPrime == true) {
```

PROBLEMS 1   OUTPUT   TERMINAL   DEBUG CONSOLE   zsh

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
3
n is prime
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.ja
4
n is not prime
shradhakhapra@Shradhas-Air Java Programming Language %
```



Run Testcases   ⊗ 0 ⚠ 1   Spaces: 4   UTF-8   LF   Java   Go Live